NYC Weather & City Traffic

A Data-Driven Model for Predicting Traffic Delays

PRESENTED BY

Linwei Zheng • Cai Rong Li • Christopher Neria • Mohamed K • Ibsan Hossain

OBJECTIVE

Build a data-driven model that uses weather conditions to analyze and predict traffic delays across New York City.

- How does rainfall affect traffic speed across different NYC boroughs?
- What times of day experience the worst traffic slowdowns?
- Where are the most congested segments in the city?
- Can we improve forecast accuracy by incorporating historical traffic patterns?

Stakeholders: Commuters • Transportation Agencies • City Planners

DATA PIPELINE

visualcrossing

NYC OpenData

Extract → **Transform** → MySQL **Load** → tableau **Visualize**

End-to-end ETL pipeline from raw API calls to interactive dashboards

DATA SOURCES

Visual Crossing

Weather API

- Temperature, rainfall, humidity
- Wind speed & snow accumulation
- 1,000 calls/day free tier

NYC OpenData

DOT Traffic Speeds

- Avg speed (mph) by segment
- Travel time + traffic volume by date
- Real-time

Real-time APIs feed our unified weather + traffic database

Regression Analysis

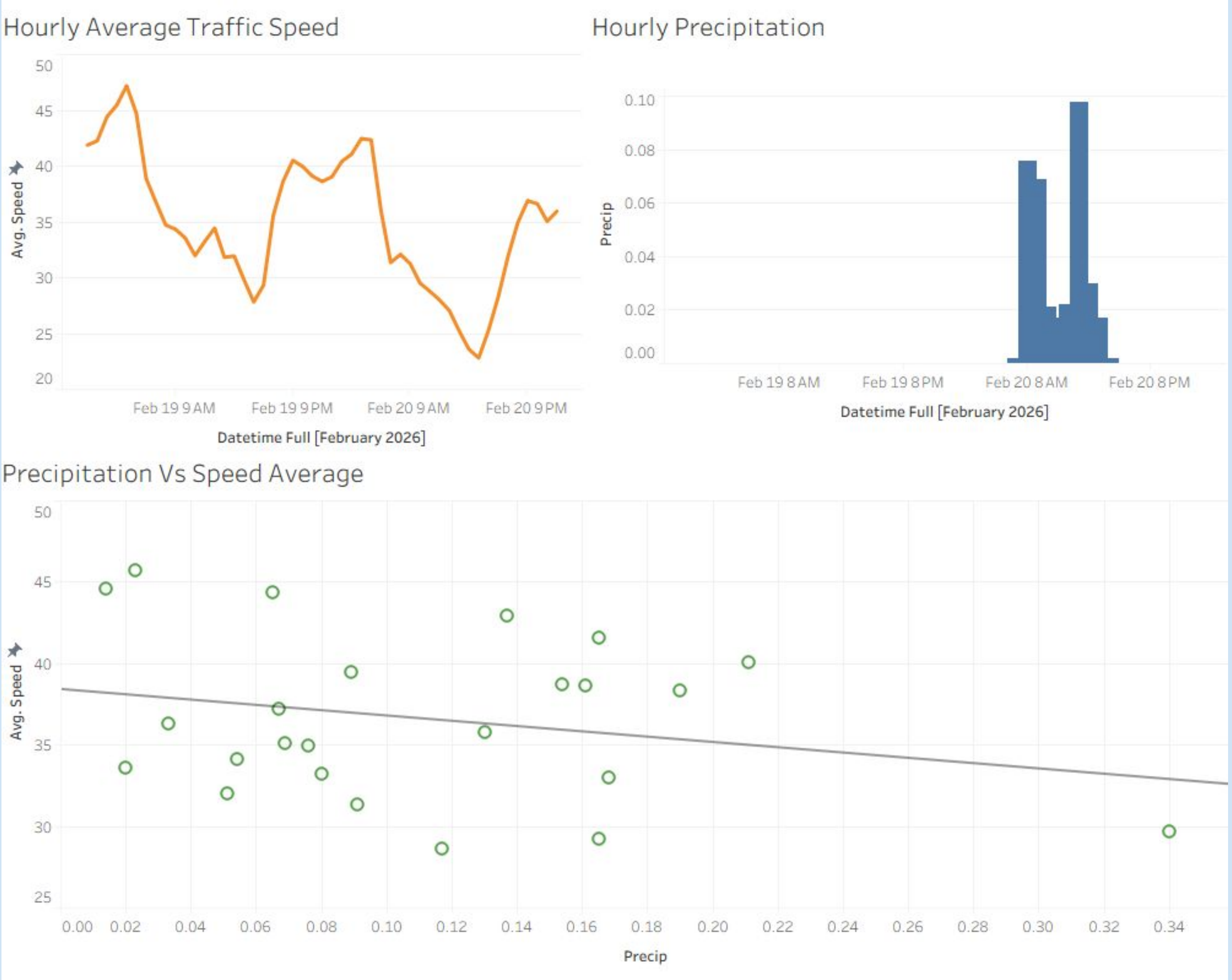
Our analysis and visualization prep strategy

WHAT OUR MODEL DOES

- Analyzes the relationship between weather conditions and NYC traffic patterns across all five boroughs
- Uses regression analysis to measure how rainfall and temperature affect average traffic speeds
- Compares different forecasting approaches to find the most accurate prediction method
- Identifies when and where traffic slowdowns are most severe
- Provides data-driven insights for commuters, transportation agencies, and city planners

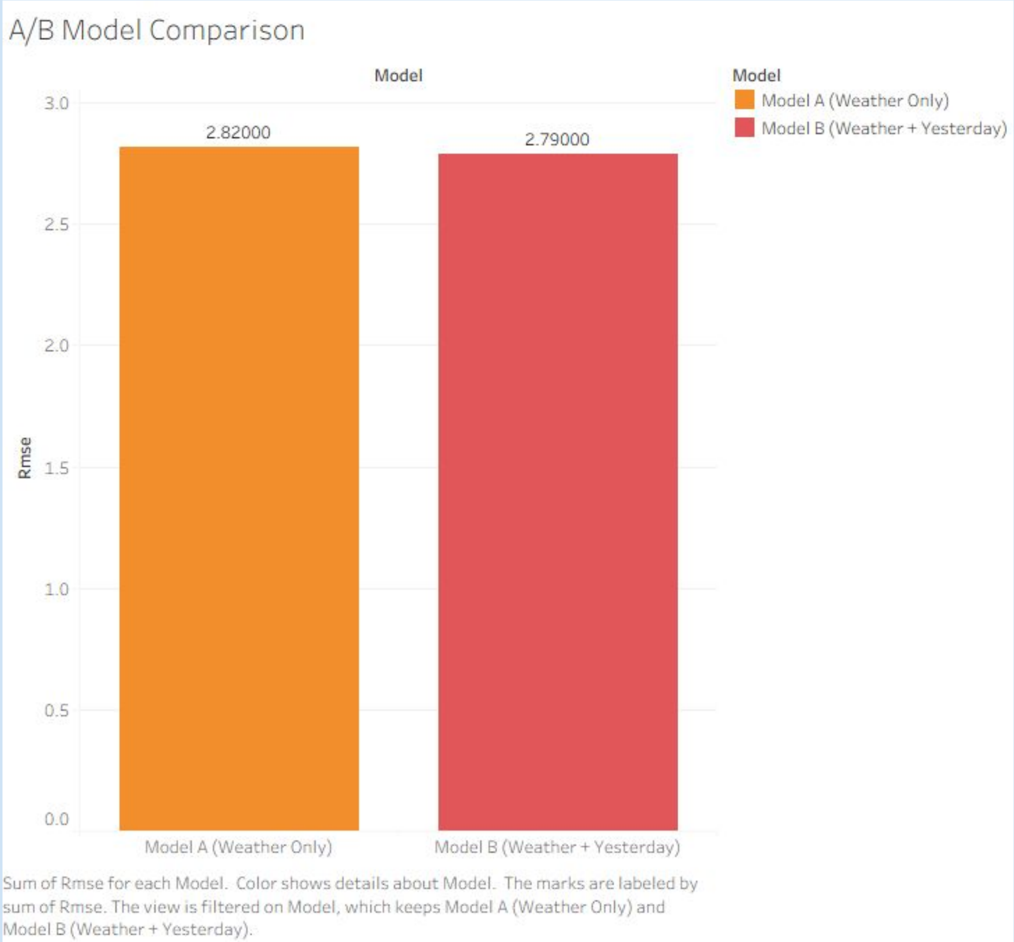
RAINFALL vs SPEED: REGRESSION FIT

Trend line suggests that for every 0.1 inch increase in rainfall, average traffic speed decreases by about **1.63 mph**.



A/B MODEL COMPARISON

Model B (Weather + Yesterday's Traffic) outperforms Model A by 29%

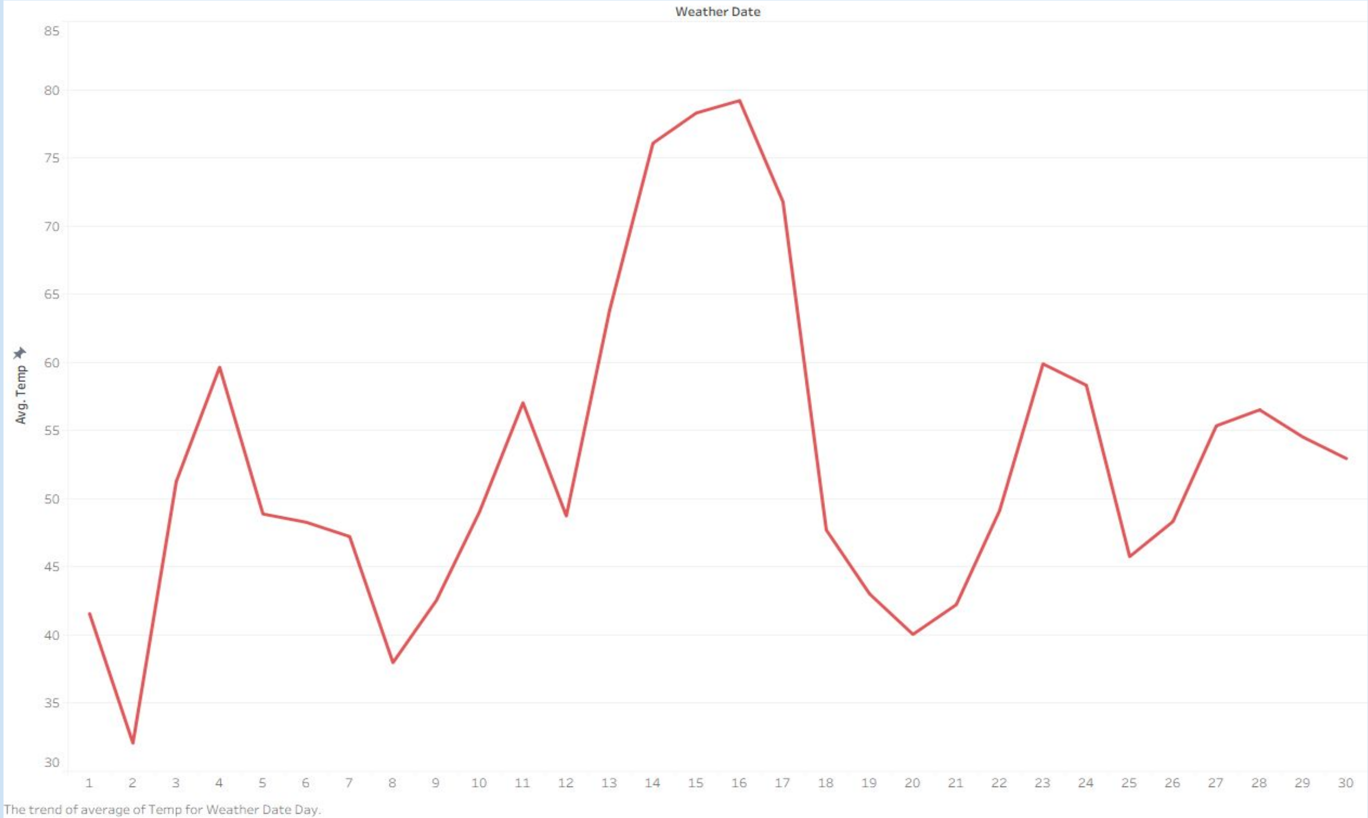


WHY THIS MATTERS

- Weather consistently impacts traffic speeds citywide, with some boroughs experiencing more significant slowdowns than others
- Understanding these patterns helps commuters plan better routes and departure times
- Transportation agencies can use these insights to allocate resources more effectively during weather events
- City planners can identify chronic problem areas that need infrastructure improvements
- Better forecasting leads to better decisions for everyone who uses NYC's roads

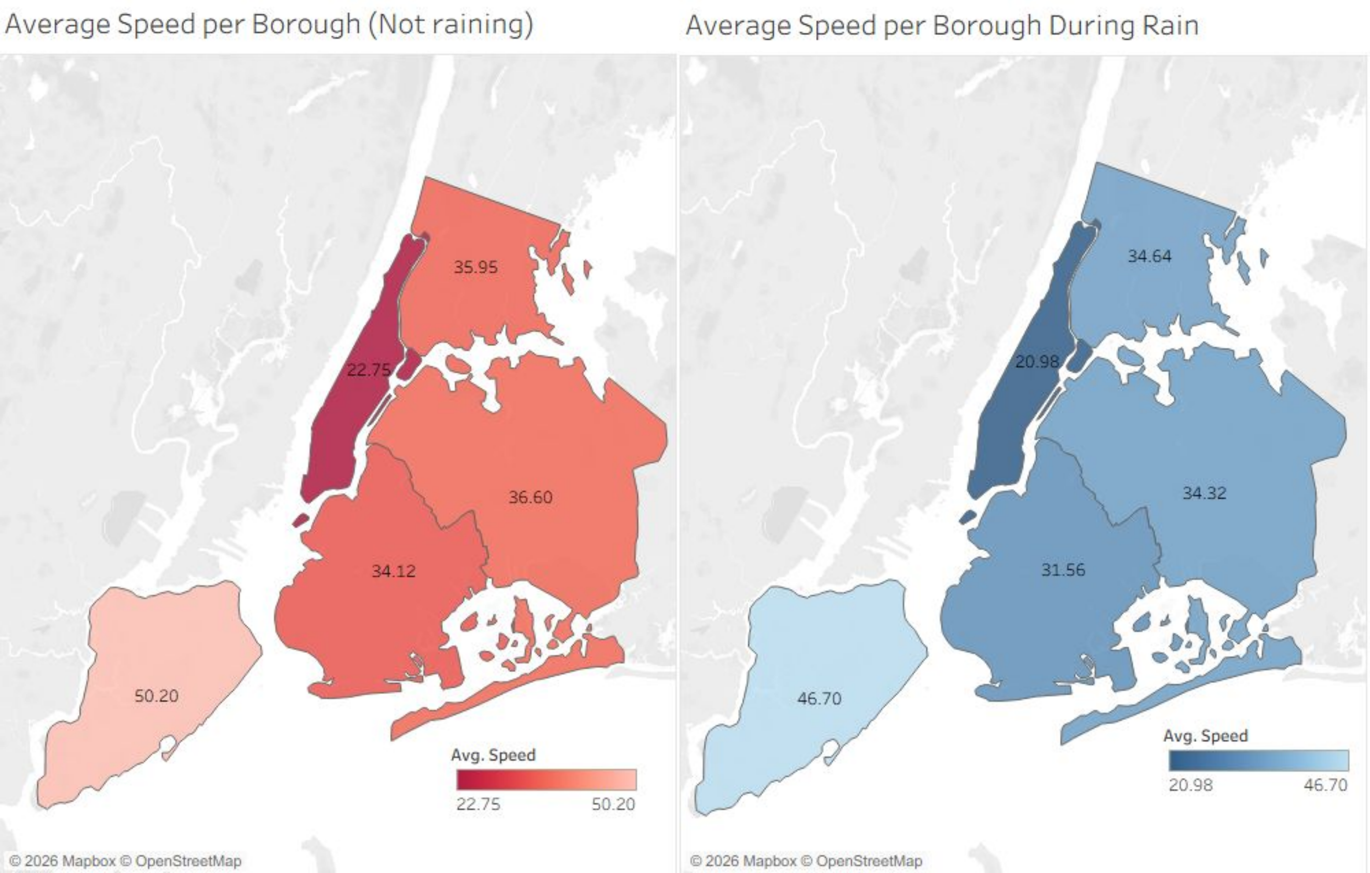
Visualizations

Average Temperature Over Study Window



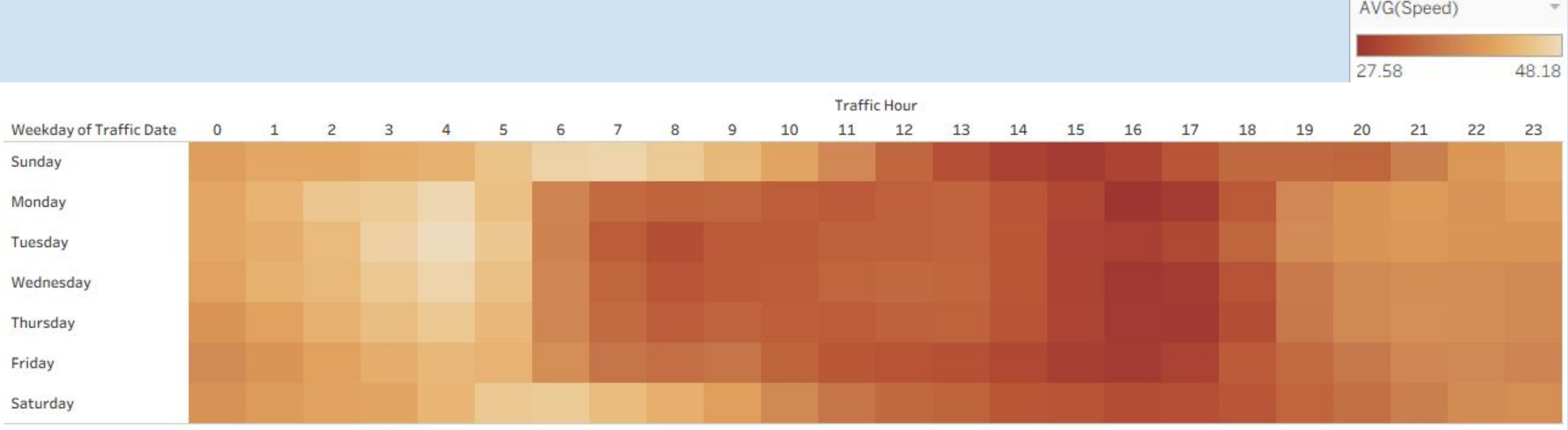
This chart shows that average temperature changes a lot during the study window, rising sharply to its highest point around days 15–16 near 79°F before dropping quickly afterward. This matters because temperature can affect travel behavior and traffic conditions, so commuters and transportation agencies should consider weather changes when analyzing congestion patterns. .

Average Speed by Borough



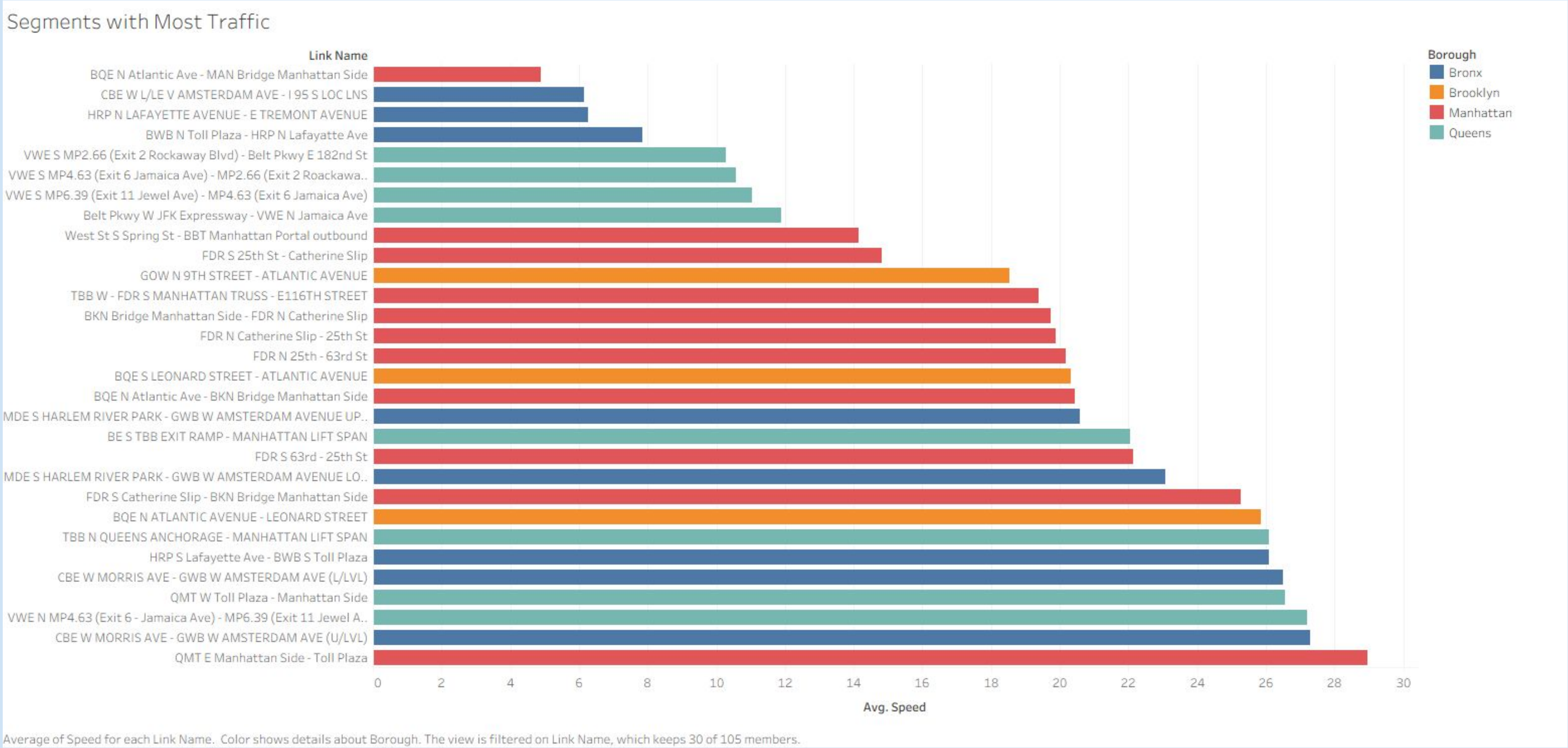
Every borough loses speed when it rains: Manhattan drops from 22.75 to 20.98 mph (-7.8%) and Brooklyn from 34.12 to 31.56 mph (-7.5%) — the largest percentage hits. For commuters this confirms rainy-day trips are slower citywide, and for DOT and planners it pinpoints the priority boroughs for wet-weather signal timing.

Speed Average by Hour of Day and Day of Week



Speeds drop on weekday afternoons, especially roughly 1–6 PM Monday through Friday, with the slowest periods near 27.6 mph, while speeds around 5–7 AM are much faster, approaching 48.2 mph. This suggests workers should leave for work around 5–7 AM if possible, while transportation agencies can target signal timing, transit support, and roadwork planning during predictable slow-speed windows.

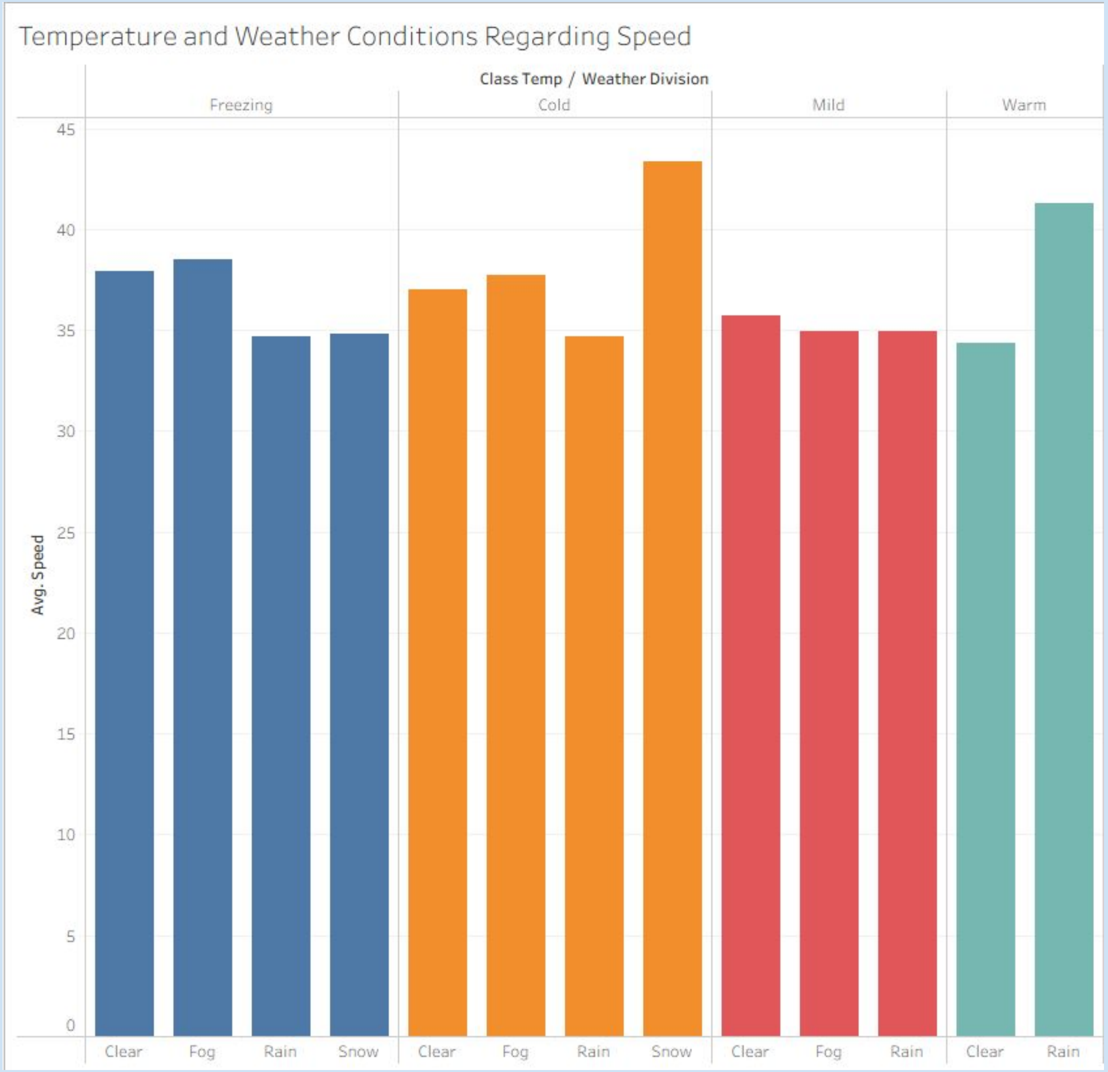
NYC Congestion Map — Avg Speed by Segment



The slowest segments include major bridge approaches and expressway corridors, with the BQE North Atlantic Avenue to Manhattan Bridge Manhattan Side segment averaging roughly 5 mph, followed by the Cross Bronx Expressway westbound near Amsterdam Avenue/I-95 south local lanes at about 6 mph. Manhattan appears most often among the listed congested segments, showing commuters which corridors to avoid when possible and giving transportation agencies a ranked list of choke points for infrastructure or traffic-operation improvements.

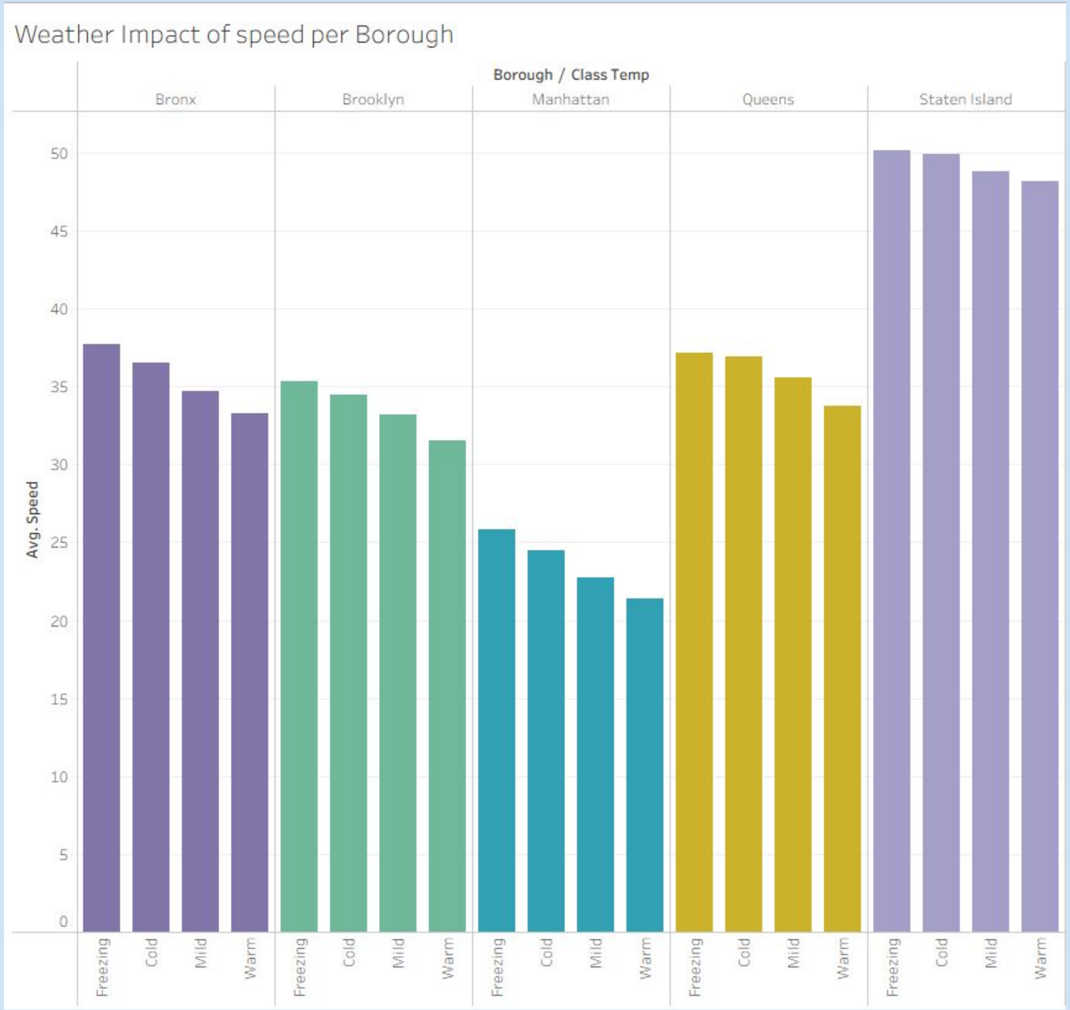
Speed Distribution by Weather Condition and Temperature

The chart suggests that some colder or worse-weather categories have faster average speeds, possibly because fewer people choose to travel in those conditions, reducing traffic volume. This is important because commuters and transportation agencies should consider that weather can change both driving conditions and the number of people on the road, which can affect average speeds.



Weather Impact on Speed by Borough

The chart shows that Staten Island has the highest average speeds across all temperature categories, while Manhattan has the lowest, suggesting Manhattan experiences the most congestion. Speeds also appear slightly faster in freezing or colder conditions, which may be because fewer people choose to travel in bad weather, reducing traffic volume and allowing vehicles to move faster.



Business Recommendations

Impact on Stakeholders

- Commuters can choose better times and routes on rainy days
- DOT can predict where to send crews based on weather forecasts
- City planners can target infrastructure spending on the slowest roads
- Emergency services can prepare for congestion hotspots

Core Findings

- Rain slows traffic everywhere, especially in Manhattan and Brooklyn
- Weekday afternoons (1–6 PM) get worse when weather is bad
- Using yesterday's traffic data improves today's predictions
- Certain road segments always move slowly regardless of weather

Immediate Actions

- Start using the weather and traffic forecast together
- Send alerts when rain hits during rush hour
- Test smarter traffic signals in Manhattan and Brooklyn

Future Enhancements

- Extend the model to cover all weather conditions year-round
- Create separate models for each borough
- Connect forecasts to navigation apps and transit alerts

Thank you

Questions?